

CE 310 — Week 14 Assignment: Bootstrap Applied to a New Building

CE 310 | Points: 60 pts (3 problems) | Individual submission | Due: Wed Dec 2, 9:00 AM — submit to D2L

The in-class exercise bootstrapped TxDOT bid tabulations. This assignment carries the same methods across to building energy — a different domain, different columns, and numbers nobody has walked you through. That transfer is the point: the capstone will ask you to apply these tools to data nobody in the class has analysed before, and the methods have to travel with you rather than staying attached to one dataset.

How to Submit

Read this before you submit. Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

Requirement	Detail
Run all cells	Runtime → Run all — wait until every cell shows a checkmark, no spinners
File format	Download as .ipynb — File → Download → Download .ipynb (not PDF, not .py)
Print lines	Do not edit or delete any <code>print(f"ANSWER_... = ...")</code> line
Seed	Do not change <code>SEED = 42</code> — results must use this exact seed
Name and NetID	Fill in NAME and NETID in the identity cell and run it — that cell is what matches your work to your student record. A blank NETID takes a 5-point deduction.
File name	CE310_W14_Assignment_<NetID>.ipynb e.g. CE310_W14_Assignment_abc123.ipynb
Submission	Upload the .ipynb file to D2L → Assignments by 9:00 AM, Wed Dec 2

Dataset

Pick any one of the five building datasets below and use it for every problem. State which one you chose in CSV_FILE in the setup cell — your numbers will differ from your classmates', so your submission has to say which building they come from.

Week15_Hospital_Dataset.csv
Week15_PrimarySchool_Dataset.csv
Week15_LargeRetail_Dataset.csv
Week15_Warehouse_Dataset.csv
Week15_SmallHotel_Dataset.csv

All five share the same column structure — Outdoor_Temp_F, Cooling_kWh, BusinessHours (Yes/No) — so once you have written the code for one building, swapping the filename is the only change needed to run it on another.

The bootstrap loops themselves are the ones from the exercise: resample rows with replacement, refit, take the 2.5th and 97.5th percentiles. What changes is the column names and what the numbers mean. If you

find yourself unable to adapt a loop, that is the signal to re-read the exercise cell it came from rather than to start over.

The five buildings behave very differently: a hospital runs around the clock, a primary school empties at weekends and all summer, a warehouse has a large insulated thermal mass, a small hotel fills up at night. **Pick one whose operation you can reason about** — Problems 2 and 3 ask you to explain the sign and size of what you find, not just report it.

These are the same five datasets offered as the ready-made option for the Week 15 capstone. If your team plans to use one, this assignment is a useful head start — but you are not committed to it either way.

A note on runtime. Every bootstrap here refits a regression a thousand times. On Colab that is a minute or two per loop, which is normal — do not assume a running cell has hung. If you want it faster, `np.linalg.lstsq` on the design matrix gives the same coefficients as `smf.ols` and is roughly twenty times quicker.

Problem 1 — Bootstrapping Your Building (30 pts)

Use `N_BOOT = 1000` and `SEED = 42` throughout unless a part says otherwise. All confidence intervals are **percentile** intervals: the 2.5th and 97.5th percentiles of the bootstrap replicates.

P1a (8 pts) — A mean and a slope

Bootstrap the **mean outdoor temperature**, and separately the **slope** of the simple model `Cooling_kWh ~ Outdoor_Temp_F`.

```
ANSWER_P1a_temp_lower = ____ # 3 decimals
ANSWER_P1a_temp_upper = ____
ANSWER_P1a_slope_lower = ____ # 4 decimals
ANSWER_P1a_slope_upper = ____
```

All five buildings share the same Tucson outdoor temperatures, so your temperature interval should match your classmates' whatever building they picked. Your slope will not.

In one sentence: compare your temperature CI to the analytical t-interval (`stats.t.interval`). Do they agree?

P1b (8 pts) — A coefficient and a fit statistic

Fit `Cooling_kWh ~ Outdoor_Temp_F + BH` and bootstrap **both** the BH coefficient and the model's R^2 , using a pairs bootstrap — resample whole rows, refit, collect.

```
ANSWER_P1b_bh_lower = ____ # 4 decimals
ANSWER_P1b_bh_upper = ____
ANSWER_P1b_r2_lower = ____
ANSWER_P1b_r2_upper = ____
```

Does your BH interval include zero? If it does, the occupancy effect is not distinguishable from noise in your building. If it does not, note the **sign** — one of the five buildings has a negative BH coefficient, and it is not a mistake.

P1c (7 pts) — Does more resampling narrow the interval?

Recompute the BH coefficient's CI **width** (upper minus lower) at `N_BOOT = 200` and at `N_BOOT = 1000`.

ANSWER_P1c_width_200 = ____ # 4 decimals
ANSWER_P1c_width_1000 = ____

In 1–2 sentences: In Week 13 the spread of a Monte Carlo estimate shrank by about a factor of 3 every time N went up tenfold. Here the interval **width** barely moves. Explain the difference — what does raising N_BOOT actually reduce, and what is the width of a bootstrap CI a property of?

P1d (7 pts) — Is the effect real, and how precise is it?

Run a **permutation test** on the BH effect: shuffle the BH column, refit, collect the coefficient, and report the two-sided p-value — the fraction of shuffled coefficients at least as large in absolute value as the one you observed. Then report the **bootstrap standard error** of the BH coefficient: the standard deviation of your P1b replicates.

ANSWER_P1d_bh_coef = ____ # the observed BH coefficient, 4 decimals
ANSWER_P1d_pvalue = ____ # 4 decimals
ANSWER_P1d_bh_se = ____ # std of the bootstrap replicates, 4 decimals

Problem 2 — Do the Two Methods Agree? (18 pts)

P2a (6 pts) — Bootstrap width against analytical width

statsmodels will give you the analytical 95% CI for the BH coefficient directly, from `model.conf_int()`. Report the ratio of your bootstrap CI width to that analytical width:

ANSWER_P2a_width_ratio = ____ # bootstrap width / analytical width, 3 decimals

A ratio near 1.0 means the two methods agree. Meaningfully above 1.0 means the bootstrap found uncertainty the formula did not.

P2b (6 pts — written)

In 3–5 sentences, interpret the BH coefficient for **your** building. State the point estimate and the interval. Does the interval include zero? Then explain the **sign and the size** in terms of how that building actually operates — a hospital's night wards, a school's empty summer, a warehouse's thermal mass, a hotel's evening occupancy. A coefficient of +12 kWh/hr and one of –3 kWh/hr are both correct answers for different buildings; say why yours is what it is.

P2c (6 pts — written)

Your P2a ratio is either close to 1.0 or clearly above it, depending on which building you chose.

In 3–5 sentences: state your ratio and say which case you are in. The analytical formula assumes the residuals have constant variance and are independent; the bootstrap assumes neither. If your ratio is near 1.0, explain what that tells you about those assumptions for your building. If it is well above 1.0, name the feature of your building's operation that plausibly breaks one of them — and say which assumption you think is failing.

Problem 3 — Reflection (12 pts)

100–150 words, typed prose, not bullet points. Cite at least two specific numbers from your own analysis.

The bootstrap replaces a formula with resampling, and in this assignment it mostly agreed with the formula. That agreement is the reason to trust it — but it is not a licence to use it everywhere.

Describe **one** situation in civil or architectural engineering where the bootstrap would give a misleading answer, and say what specifically goes wrong. Your hourly data is a live example to think with: consecutive hours are strongly correlated, and a pairs bootstrap that resamples individual rows treats them as independent. Does that matter for what you estimated here? Would it matter more for something else?

Grading Summary

Problem	Description	Points
P1a	Bootstrap CI for the mean temperature and the simple-model slope	8
P1b	Bootstrap CI for the BH coefficient and R^2	8
P1c	CI width at two resample counts, and why it does not shrink	7
P1d	Permutation p-value and the bootstrap standard error	7
P2a	Bootstrap CI width ÷ analytical CI width	6
P2b	Written: the BH coefficient explained from how the building runs	6
P2c	Written: which assumption the ratio is testing, and whether it holds	6
P3	Reflection: where the bootstrap misleads	12
Total		60

Connection to Week 15

Everything here — a CI you can defend, an effect you have tested two ways, and a clear statement of what your method assumes — is what the capstone report asks for. The building you chose is one you may keep.

Deliverable

Submit a single Colab notebook (.ipynb) with all cells executed. Every ANSWER_ variable must appear in cell output. Written responses go in markdown cells immediately after the code cell they discuss.